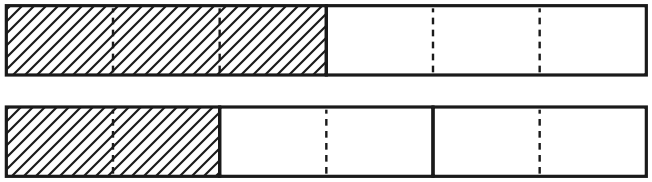
Unlike Fractions

Different bottoms? Cut the pieces until they match, then add or take away.

1/2 + 1/3: halves and thirds don't match



Cut each half into 3: sixths.
Cut each third into 2: sixths.
Now the pieces match.

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

Find the same size pieces

Count by the bigger bottom until the other bottom fits: 4, 8, 12. 3 fits in 12.

$$\frac{2}{3} - \frac{1}{4} = \frac{8}{12} - \frac{3}{12} = \frac{5}{12}$$

Mixed numbers: wholes, then pieces

$$1\frac{1}{2} + 2\frac{1}{3} = 3\frac{5}{6}$$

1 + 2 = 3 wholes.
1/2 + 1/3 = 5/6.

Your turn!

$$\frac{1}{4} + \frac{1}{6} = \frac{\boxed{3}}{12} + \frac{\boxed{2}}{12} = \frac{\boxed{5}}{12}$$

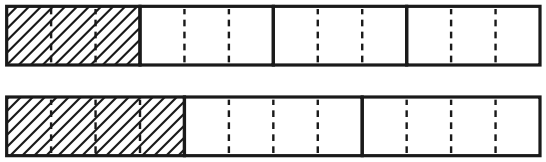
For teachers

Goal: add and subtract fractions with different bottoms by making the same size pieces (5.NF.1–2). It uses packet 61's equal fractions. Say “We can only add pieces of the same size.” Any common bottom is fine.

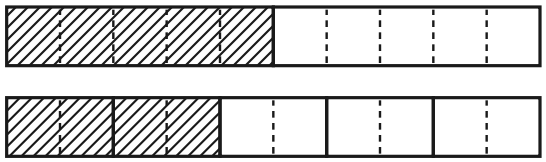
Unlike Fractions: Practice

Same size pieces first. Then add or take away the tops.

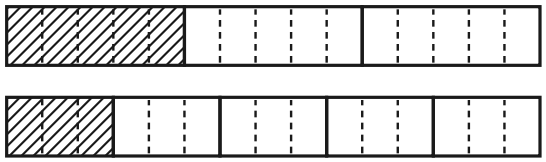
Part 1 Cut the pieces The dashed lines make the same size pieces.

1) 

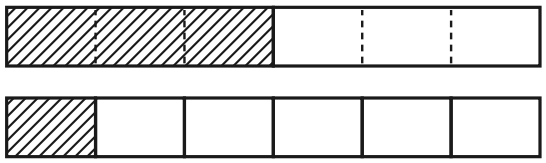
$$\frac{1}{4} + \frac{1}{3} = \frac{\boxed{3}}{12} + \frac{\boxed{4}}{12} = \frac{\boxed{7}}{12}$$

2) 

$$\frac{1}{2} + \frac{2}{5} = \frac{\boxed{5}}{10} + \frac{\boxed{4}}{10} = \frac{\boxed{9}}{10}$$

3) 

$$\frac{1}{3} + \frac{1}{5} = \frac{\boxed{5}}{15} + \frac{\boxed{3}}{15} = \frac{\boxed{8}}{15}$$

4) 

$$\frac{1}{2} + \frac{1}{6} = \frac{\boxed{3}}{6} + \frac{\boxed{1}}{6} = \frac{\boxed{4}}{6}$$

Part 2 Same bottoms The new bottom is given.

5) $\frac{1}{3} + \frac{3}{4} = \frac{\boxed{4}}{12} + \frac{\boxed{9}}{12} = \frac{\boxed{13}}{12}$

6) $\frac{1}{2} + \frac{3}{8} = \frac{\boxed{4}}{8} + \frac{\boxed{3}}{8} = \frac{\boxed{7}}{8}$

7) $\frac{5}{6} - \frac{1}{3} = \frac{\boxed{5}}{6} - \frac{\boxed{2}}{6} = \frac{\boxed{3}}{6}$

Unlike Fractions: Practice (continued)

Part 2 Same bottoms (continued)

8) $\frac{2}{5} + \frac{1}{4} = \frac{\boxed{8}}{20} + \frac{\boxed{5}}{20} = \frac{\boxed{13}}{20}$

9) $\frac{7}{10} - \frac{1}{2} = \frac{\boxed{7}}{10} - \frac{\boxed{5}}{10} = \frac{\boxed{2}}{10}$

10) $\frac{1}{6} + \frac{4}{9} = \frac{\boxed{3}}{18} + \frac{\boxed{8}}{18} = \frac{\boxed{11}}{18}$

11) $\frac{3}{4} + \frac{1}{10} = \frac{\boxed{15}}{20} + \frac{\boxed{2}}{20} = \frac{\boxed{17}}{20}$

12) $\frac{5}{8} - \frac{1}{4} = \frac{\boxed{5}}{8} - \frac{\boxed{2}}{8} = \frac{\boxed{3}}{8}$

Part 3 Add No help this time. Find the same size pieces.

13) $\frac{1}{2} + \frac{1}{5} = \frac{\boxed{7}}{\boxed{10}}$

14) $\frac{2}{3} + \frac{1}{4} = \frac{\boxed{11}}{\boxed{12}}$

15) $\frac{1}{4} + \frac{3}{5} = \frac{\boxed{17}}{\boxed{20}}$

16) $\frac{3}{8} + \frac{1}{3} = \frac{\boxed{17}}{\boxed{24}}$

17) $\frac{1}{6} + \frac{2}{3} = \frac{\boxed{5}}{\boxed{6}}$

18) $\frac{2}{5} + \frac{1}{2} = \frac{\boxed{9}}{\boxed{10}}$

Unlike Fractions: Practice (continued)

Part 3 Add (continued)

19) $\frac{1}{2} + \frac{3}{7} = \frac{13}{14}$

20) $\frac{3}{4} + \frac{2}{3} = \frac{17}{12}$

21) $\frac{1}{3} + \frac{2}{9} = \frac{5}{9}$

22) $\frac{5}{6} + \frac{1}{8} = \frac{23}{24}$

23) $\frac{3}{10} + \frac{1}{4} = \frac{11}{20}$

24) $\frac{2}{7} + \frac{1}{3} = \frac{13}{21}$

25) $\frac{1}{5} + \frac{3}{8} = \frac{23}{40}$

26) $\frac{4}{9} + \frac{1}{6} = \frac{11}{18}$

27) $\frac{1}{8} + \frac{5}{12} = \frac{13}{24}$

28) $\frac{3}{5} + \frac{1}{6} = \frac{23}{30}$

Unlike Fractions: Practice (continued)

Part 4 Subtract No help this time. Find the same size pieces.

29) $\frac{3}{4} - \frac{1}{3} = \frac{5}{12}$

30) $\frac{5}{6} - \frac{1}{4} = \frac{7}{12}$

31) $\frac{2}{3} - \frac{1}{5} = \frac{7}{15}$

32) $\frac{7}{8} - \frac{1}{6} = \frac{17}{24}$

33) $\frac{4}{5} - \frac{1}{2} = \frac{3}{10}$

34) $\frac{5}{8} - \frac{1}{3} = \frac{7}{24}$

35) $\frac{9}{10} - \frac{1}{4} = \frac{13}{20}$

36) $\frac{3}{4} - \frac{2}{7} = \frac{13}{28}$

37) $\frac{5}{6} - \frac{2}{9} = \frac{11}{18}$

38) $\frac{2}{3} - \frac{1}{8} = \frac{13}{24}$

39) $\frac{7}{10} - \frac{1}{3} = \frac{11}{30}$

40) $\frac{4}{9} - \frac{1}{6} = \frac{5}{18}$

Unlike Fractions: Practice (continued)

Part 4 Subtract (continued)

41) $\frac{7}{12} - \frac{1}{8} = \frac{11}{24}$

42) $\frac{3}{5} - \frac{1}{4} = \frac{7}{20}$

43) $\frac{6}{7} - \frac{1}{2} = \frac{5}{14}$

44) $\frac{5}{9} - \frac{1}{6} = \frac{7}{18}$

Part 5 Mixed numbers No help this time. Wholes, then pieces.

45) $1\frac{1}{2} + 2\frac{1}{4} = 3\frac{3}{4}$

46) $2\frac{1}{3} + 1\frac{1}{2} = 3\frac{5}{6}$

47) $3\frac{1}{4} + 1\frac{2}{3} = 4\frac{11}{12}$

48) $1\frac{2}{5} + 2\frac{1}{2} = 3\frac{9}{10}$

49) $4\frac{1}{6} + 1\frac{1}{4} = 5\frac{5}{12}$

50) $3\frac{2}{3} - 1\frac{1}{2} = 2\frac{1}{6}$

51) $5\frac{3}{4} - 2\frac{1}{3} = 3\frac{5}{12}$

52) $4\frac{5}{6} - 1\frac{1}{4} = 3\frac{7}{12}$

Unlike Fractions: Practice (continued)

Part 5 Mixed numbers (continued)

53) $6\frac{1}{2} - 3\frac{1}{5} = 3\frac{3}{10}$

Part 6 Add

54) $\frac{3}{5} + \frac{1}{2} = \frac{11}{10}$

55) $\frac{2}{4} + \frac{1}{3} = \frac{10}{12}$

56) $\frac{5}{6} + \frac{5}{8} = \frac{35}{24}$

57) $\frac{2}{4} + \frac{4}{5} = \frac{26}{20}$

58) $\frac{1}{6} + \frac{1}{4} = \frac{5}{12}$

59) $\frac{6}{8} + \frac{2}{5} = \frac{46}{40}$

60) $\frac{1}{3} + \frac{1}{5} = \frac{8}{15}$

61) $\frac{3}{4} + \frac{2}{5} = \frac{23}{20}$

62) $\frac{1}{8} + \frac{2}{3} = \frac{19}{24}$

63) $\frac{2}{8} + \frac{2}{5} = \frac{26}{40}$

Unlike Fractions: Practice (continued)

Part 6 Add (continued)

64) $\frac{1}{5} + \frac{1}{8} = \frac{13}{40}$

65) $\frac{2}{3} + \frac{3}{4} = \frac{17}{12}$

66) $\frac{1}{4} + \frac{2}{3} = \frac{11}{12}$

67) $\frac{3}{4} + \frac{1}{3} = \frac{13}{12}$

68) $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$

69) $\frac{3}{6} + \frac{3}{4} = \frac{15}{12}$

70) $\frac{2}{3} + \frac{1}{2} = \frac{7}{6}$

71) $\frac{3}{6} + \frac{2}{5} = \frac{27}{30}$

Part 7 Subtract

72) $\frac{3}{4} - \frac{4}{6} = \frac{1}{12}$

73) $\frac{4}{5} - \frac{2}{4} = \frac{6}{20}$

74) $\frac{3}{5} - \frac{1}{3} = \frac{4}{15}$

75) $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

Unlike Fractions: Practice (continued)

Part 7 Subtract (continued)

76) $\frac{2}{3} - \frac{3}{5} = \frac{1}{15}$

77) $\frac{4}{5} - \frac{1}{8} = \frac{27}{40}$

78) $\frac{2}{3} - \frac{1}{4} = \frac{5}{12}$

79) $\frac{3}{4} - \frac{2}{5} = \frac{7}{20}$

80) $\frac{2}{4} - \frac{1}{6} = \frac{4}{12}$

81) $\frac{1}{4} - \frac{1}{5} = \frac{1}{20}$

82) $\frac{3}{6} - \frac{1}{5} = \frac{9}{30}$

83) $\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$

84) $\frac{2}{5} - \frac{1}{6} = \frac{7}{30}$

85) $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$

86) $\frac{5}{6} - \frac{2}{4} = \frac{4}{12}$

87) $\frac{3}{8} - \frac{1}{6} = \frac{5}{24}$

Unlike Fractions: Practice (continued)

Part 7 Subtract (continued)

88) $\frac{1}{2} - \frac{2}{5} = \frac{1}{10}$

89) $\frac{6}{8} - \frac{4}{6} = \frac{2}{24}$

Part 8 Remember? Problems from earlier packets.

90) $7 \times 0.6 = 4.2$

91) $3.6 \div 4 = 0.9$

92) $0.7 \times 0.3 = 0.21$

93) $945 \div 35 = 27$

94) $\frac{5}{6} = \frac{15}{18}$

95) $\frac{3}{8} < \frac{1}{2}$

Part 9 Word problems Same size pieces first.

96) Mia walked 1/2 mile to school and 1/3 mile to the park. How far did she walk?

Number sentence **1/2 + 1/3 = 5/6**

Answer **5/6** mile

Draw or show your work

97) A jug had 3/4 L of milk. Ben drank 1/3 L. How much milk is left?

Number sentence **3/4 − 1/3 = 5/12**

Answer **5/12** L

Draw or show your work

98) Leo read 2/5 of a book on Monday and 1/4 of it on Tuesday. How much did he read?

Number sentence **2/5 + 1/4 = 13/20**

Answer **13/20** of the book

Draw or show your work

Unlike Fractions: Practice (continued)

Part 9 Word problems (continued)

99) A board is 2 1/2 m long. Zoe cuts off 1 1/3 m. How long is the rest?

Number sentence **2 1/2 − 1 1/3 = 1 1/6**

Answer **1 1/6** m

Draw or show your work

100) Challenge: Ava had 1 whole pizza. She ate 1/4 and Sam ate 1/3. How much is left?

Number sentence **1/4 + 1/3 = 7/12, 1 − 7/12 = 5/12**

Answer **5/12** of the pizza

Draw or show your work

Check-In: Unlike Fractions

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

Part 1 Same bottoms The new bottom is given.

1) $\frac{3}{8} + \frac{1}{6} = \frac{9}{24} + \frac{4}{24} = \frac{13}{24}$

Part 2 Add

2) $\frac{1}{2} + \frac{1}{7} = \frac{9}{14}$

3) $\frac{3}{4} + \frac{1}{5} = \frac{19}{20}$

4) $\frac{1}{6} + \frac{2}{5} = \frac{17}{30}$

5) $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$

Part 3 Subtract

6) $\frac{5}{8} - \frac{1}{6} = \frac{11}{24}$

7) $\frac{4}{5} - \frac{1}{3} = \frac{7}{15}$

8) $\frac{3}{4} - \frac{1}{5} = \frac{11}{20}$

9) $\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$

Part 4 Mixed numbers

10) $4\frac{2}{3} - 2\frac{1}{4} = 2\frac{5}{12}$

11) $1\frac{1}{6} + 3\frac{1}{4} = 4\frac{5}{12}$